Phase 4 Derivation Draft
Choice Regions and the Incumbent Transition Matrix

1 Choice Regions and the Incumbent Transition Matrix

This phase studies how the solved Bellman objects from Phase 3 generate benchmark choice regions and, conditional on candidate stationary incumbent measures, generate the incumbent reorganization statistics summarized by the transition matrix. Throughout this section, (w, p_m, M) remain fixed. The value-function triple from Phase 3 is taken as given, and the stationary measures μ_A, μ_B, μ_C are treated as candidate incumbent measures. Their existence and invariance are not proved here; those questions are deferred to the later stationary-distribution phase.

1.1 Benchmark Fidelity Check

Step 1: exact benchmark objects relevant for the present task. The authoritative benchmark sources imply that the relevant Phase 4 objects are:

- the benchmark choice regions

  $$\mathcal{Z}_{sj}(M) = \{z \in Z : j \text{ solves the Bellman problem for a current type-}s \text{ firm at state } z\},$$

  where $s \in \{A, B, C\}$ and $j \in \{A, B, C, \text{exit}\}$;
- the candidate stationary incumbent measures μ_A, μ_B, μ_C;
- the benchmark transition probabilities

  $$P_{sj}(M) = \frac{\mu_s\big(\mathcal{Z}_{sj}(M)\big)}{\mu_s(Z)};$$
- the incumbent transition matrix $\mathbf{P}(M)$ with three current-type rows and four chosen-action columns;
- the organizational policy rule $g_s(z; M)$ that appears later in the stationary-measure law of motion.

These objects are not all of the same kind. The sets $\mathcal{Z}_{sj}(M)$ are derived objects built from the Bellman problem. The measures μ_s are endogenous stationary objects that will be solved later. The transition probabilities $P_{sj}(M)$ are derived statistics formed from those two ingredients.

Step 2: exact formulas, argument lists, and interpretation. The benchmark manuscript writes the choice regions exactly as

$$\mathcal{Z}_{sj}(M) = \{z \in Z : j \text{ solves the Bellman problem for a current type-}s \text{ firm at state } z\}.$$

It then defines

$$P_{sj}(M) = \frac{\mu_s\big(\mathcal{Z}_{sj}(M)\big)}{\mu_s(Z)}.$$

Economically, $P_{sj}(M)$ is intended to describe the model-implied probability that an incumbent currently of type s reorganizes into current-period organizational form j in the stationary cross section. The benchmark manuscript further stacks these objects into the matrix

$$\mathbf{P}(M) = \begin{pmatrix} P_{AA}(M) & P_{AB}(M) & P_{AC}(M) & P_{A,\text{exit}}(M) \\ P_{BA}(M) & P_{BB}(M) & P_{BC}(M) & P_{B,\text{exit}}(M) \\ P_{CA}(M) & P_{CB}(M) & P_{CC}(M) & P_{C,\text{exit}}(M) \end{pmatrix}.$$

Later, the benchmark stationary-measure law of motion uses the single-valued organizational policy rule $g_s(z; M)$ through indicators of the form $\mathbf{1}\{g_s(z; M) = j\}$.

Step 3: whether the present task can be completed faithfully from the benchmark as written. The answer is partly yes and partly no.

1. We can faithfully define the benchmark choice regions exactly as written.
2. We can faithfully prove that these choice regions are Borel measurable.
3. Conditional on given finite Borel measures μ_s with $\mu_s(Z) > 0$, we can faithfully define the ratios $\mu_s(\mathcal{Z}_{sj}(M))/\mu_s(Z)$ and show that they are well defined.
4. We cannot, however, derive the row-sum-equals-one statement from the set-based definition alone unless we add an extra no-ties condition or an explicit selector. That obstruction must be stated openly.

Step 4: inconsistency or ambiguity that must be flagged before proceeding. There is one benchmark tension that directly affects the Phase 4 derivation.

The benchmark defines transition probabilities using the solving regions $\mathcal{Z}_{sj}(M)$, but later defines the stationary law of motion using a single-valued policy rule $g_s(z; M)$. These two formulations coincide only if ties are absent almost everywhere or if an explicit tie-breaking convention has already been added. If ties occur on a set of positive μ_s-measure, then the solving regions can overlap. In that case the set-based formula

$$P_{sj}(M) = \frac{\mu_s\big(\mathcal{Z}_{sj}(M)\big)}{\mu_s(Z)}$$

need not generate rows that sum to one, because overlapping sets are counted more than once. Therefore the statement "each row sums to one" cannot be derived from the benchmark choice-region formula alone without an extra condition. We do not silently impose that extra condition. Instead we derive exactly what follows from the benchmark formula, and then state explicitly what additional assumption or convention is needed to recover a row-stochastic transition matrix.

1.2 Part 1: Benchmark Choice Regions

Benchmark object restatement. Fix a current organizational type $s \in \{A, B, C\}$ and an action $j \in \{A, B, C, \text{exit}\}$. The benchmark choice region is

$$\mathcal{Z}_{sj}(M) = \{z \in Z : j \text{ solves the Bellman problem for a current type-}s \text{ firm at state } z\}.$$

This is a derived object. It is built from the solved Bellman system of Phase 3, not from a new primitive optimization problem introduced in Phase 4.

Economic role. The set $\mathcal{Z}_{sj}(M)$ tells us which states lead a current type-s incumbent to regard action j as optimal. In words, it is the state region in which a firm of current type s chooses to remain, switch, or exit through action j. These regions are the bridge from value functions to observable reorganization patterns. They translate the dynamic optimization problem into objects that can be measured in the stationary cross section.

Mathematical problem. By Phase 3, for each current type s and each feasible action j, the choice-specific value function $W_{sj}(z; M)$ is already defined and continuous in z. Therefore saying that action j "solves the Bellman problem" means exactly that

$$W_{sj}(z; M) \geq W_{s\ell}(z; M) \qquad \text{for every } \ell \in \{A, B, C, \text{exit}\}.$$

So the benchmark choice region can be rewritten as

$$\mathcal{Z}_{sj}(M) = \bigcap_{\ell \in \{A,B,C,\text{exit}\}} \{z \in Z : W_{sj}(z; M) \geq W_{s\ell}(z; M)\}.$$

Admissible set. For every current type s, the Bellman problem compares exactly four feasible actions:

$$\mathcal{A} = \{A, B, C, \text{exit}\}.$$

Because the action set is finite, the Bellman maximum is always attained once the four choice-specific values are finite real numbers. Phase 3 already established that property. Therefore, for every $z \in Z$, at least one action solves the Bellman problem. This implies

$$\bigcup_{j \in \mathcal{A}} \mathcal{Z}_{sj}(M) = Z.$$

The union need not be disjoint, because ties can make two or more actions simultaneously optimal.

Assumptions used. The only ingredients needed here are those already established in Phase 3:

1. the state space Z is a compact metric space;
2. the fixed-price value-function triple V^* exists and is unique;

3. each benchmark choice-specific value $W_{sj}(\cdot; M)$ is continuous.

Proposition 1 (Measurability of benchmark choice regions)**.** *For every current type $s \in \{A, B, C\}$ and action $j \in \{A, B, C, \text{exit}\}$, the benchmark choice region $\mathcal{Z}_{sj}(M)$ is a Borel subset of Z.*

Proof. **Step 1: rewrite the region as a finite intersection.** By the definition of "solves the Bellman problem,"

$$\mathcal{Z}_{sj}(M) = \bigcap_{\ell \in \mathcal{A}} \{z \in Z : W_{sj}(z; M) \geq W_{s\ell}(z; M)\}.$$

So it is enough to prove that each comparison set is Borel measurable.

Step 2: study one comparison set at a time. Fix a competitor $\ell \in \mathcal{A}$. Define the difference function

$$D_{sj\ell}(z) := W_{sj}(z; M) - W_{s\ell}(z; M).$$

Phase 3 established continuity of both $W_{sj}(\cdot; M)$ and $W_{s\ell}(\cdot; M)$. The difference of two continuous functions is continuous. Therefore $D_{sj\ell} : Z \to \mathbb{R}$ is continuous.

Step 3: use the preimage characterization of closed sets. The comparison set can be written as

$$\{z \in Z : W_{sj}(z; M) \geq W_{s\ell}(z; M)\} = D_{sj\ell}^{-1}\big([0, \infty)\big).$$

The set $[0, \infty)$ is closed in $\mathbb{R}$. The preimage of a closed set under a continuous map is closed. Hence each comparison set is closed in Z.

Step 4: intersect finitely many closed sets. The set $\mathcal{Z}_{sj}(M)$ is the intersection of finitely many closed sets, one for each action $\ell \in \mathcal{A}$. A finite intersection of closed sets is closed. Because Z is a metric space, every closed subset of Z is a Borel set. Therefore $\mathcal{Z}_{sj}(M)$ is Borel measurable. □

1.3 Part 2: Benchmark Transition Probabilities Conditional on Candidate Measures

Benchmark object restatement. Now fix a current type $s \in \{A, B, C\}$ and action $j \in \{A, B, C, \text{exit}\}$. The benchmark transition probability is

$$P_{sj}(M) = \frac{\mu_s\big(\mathcal{Z}_{sj}(M)\big)}{\mu_s(Z)}.$$

This object is derived from two ingredients:

1. the choice region $\mathcal{Z}_{sj}(M)$, which tells us where action j is optimal for a current type-s firm;

2. the candidate stationary incumbent measure μ_s, which tells us how current type-s firms are distributed over states in the stationary cross section.

Economic role. The ratio above is meant to summarize the stationary fraction of current type-s incumbents whose optimal organizational action is j. This is why the benchmark treats $\mathbf{P}(M)$ as the theoretical counterpart of the empirical transition matrix estimated from panel data.

Mathematical problem. At this phase we do not yet prove existence of the measures μ_s. We therefore work conditionally on given finite Borel measures μ_A, μ_B, μ_C on Z. The only positivity condition needed to define the ratios is

$$\mu_s(Z) > 0 \qquad \text{for each current type } s.$$

Assumptions used. For the present derivation we assume:

1. each μ_s is a finite Borel measure on Z;
2. $\mu_s(Z) > 0$ for the current type under consideration;
3. the benchmark choice regions are Borel measurable, as proved above.

Proposition 2 (Well-definedness of the benchmark ratios)**.** *Fix $s \in \{A, B, C\}$ and $j \in \{A, B, C, \text{exit}\}$. If μ_s is a finite Borel measure on Z with $\mu_s(Z) > 0$, then $P_{sj}(M)$ is well defined and satisfies*

$$0 \leq P_{sj}(M) \leq 1.$$

Proof. Because $\mathcal{Z}_{sj}(M)$ is Borel measurable, the value $\mu_s(\mathcal{Z}_{sj}(M))$ is well defined. Because $\mathcal{Z}_{sj}(M) \subseteq Z$ and μ_s is a finite measure,

$$0 \leq \mu_s\big(\mathcal{Z}_{sj}(M)\big) \leq \mu_s(Z) < \infty.$$

Because the denominator $\mu_s(Z)$ is strictly positive, the ratio

$$P_{sj}(M) = \frac{\mu_s\big(\mathcal{Z}_{sj}(M)\big)}{\mu_s(Z)}$$

is a well-defined real number. Dividing the previous inequalities by $\mu_s(Z)$ yields

$$0 \leq P_{sj}(M) \leq 1.$$

□

1.4 Part 3: What the Benchmark Formula Does and Does Not Imply

Benchmark object restatement. The benchmark manuscript proceeds from the set-based definition of $P_{sj}(M)$ to the matrix $\mathbf{P}(M)$ and states that each row sums to one. We now check exactly when that statement is mathematically justified.

Economic role. This issue matters because a transition matrix is meant to describe a probability distribution over next-period organizational outcomes conditional on the current type. If a row does not sum to one, it is not a probability distribution. Therefore the row-sum question is not a minor technical detail. It determines whether the set-based benchmark object is already a genuine transition matrix or only an intermediate statistic.

Proposition 3 (Row-sum obstruction under overlapping choice regions)**.** *Fix a current type $s \in \{A, B, C\}$ and suppose μ_s is a finite Borel measure with $\mu_s(Z) > 0$. Then:*

1. *the benchmark set-based probabilities satisfy*

$$\sum_{j \in \mathcal{A}} P_{sj}(M) \geq 1;$$

2. *equality holds if and only if*

$$\sum_{j \in \mathcal{A}} \mathbf{1}\{z \in \mathcal{Z}_{sj}(M)\} = 1 \qquad \text{for } \mu_s\text{-almost every } z;$$

3. *if there exists a positive-μ_s set of ties, meaning that for some distinct $j, \ell \in \mathcal{A}$,*

$$\mu_s\big(\mathcal{Z}_{sj}(M) \cap \mathcal{Z}_{s\ell}(M)\big) > 0,$$

then

$$\sum_{j \in \mathcal{A}} P_{sj}(M) > 1.$$

Proof. **Step 1: translate the row sum into an integral of indicators.** By definition,

$$\begin{aligned}
\sum_{j \in \mathcal{A}} P_{sj}(M) &= \sum_{j \in \mathcal{A}} \frac{\mu_s\big(\mathcal{Z}_{sj}(M)\big)}{\mu_s(Z)} \\
&= \frac{1}{\mu_s(Z)} \sum_{j \in \mathcal{A}} \int_Z \mathbf{1}\{z \in \mathcal{Z}_{sj}(M)\}\, \mu_s(dz) \\
&= \frac{1}{\mu_s(Z)} \int_Z \sum_{j \in \mathcal{A}} \mathbf{1}\{z \in \mathcal{Z}_{sj}(M)\}\, \mu_s(dz).
\end{aligned}$$

Step 2: use the fact that some action is always optimal. Phase 3 established that the Bellman maximum is attained at every state because the action set $\mathcal{A}$ is finite and all four choice-specific values are finite. Therefore for every $z \in Z$, at least one indicator inside the sum equals one. Hence

$$\sum_{j \in \mathcal{A}} \mathbf{1}\{z \in \mathcal{Z}_{sj}(M)\} \geq 1 \qquad \text{for every } z \in Z.$$

Integrating both sides with respect to μ_s gives

$$\int_Z \sum_{j \in \mathcal{A}} \mathbf{1}\{z \in \mathcal{Z}_{sj}(M)\}\, \mu_s(dz) \geq \int_Z 1\, \mu_s(dz) = \mu_s(Z).$$

Dividing by $\mu_s(Z) > 0$ yields

$$\sum_{j \in \mathcal{A}} P_{sj}(M) \geq 1.$$

Step 3: characterize when equality holds. The inequality becomes equality if and only if the integrand

$$\sum_{j \in \mathcal{A}} \mathbf{1}\{z \in \mathcal{Z}_{sj}(M)\}$$

equals one for μ_s-almost every z. Because at least one action is always optimal, this is equivalent to requiring that exactly one action be counted almost everywhere. In other words, the benchmark choice regions must form a partition of Z up to μ_s-null sets.

Step 4: show why positive-measure ties force strict inequality. Suppose there are distinct actions j and ℓ such that

$$\mu_s\big(\mathcal{Z}_{sj}(M) \cap \mathcal{Z}_{s\ell}(M)\big) > 0.$$

On that overlap set, at least two indicators equal one, so

$$\sum_{r \in \mathcal{A}} \mathbf{1}\{z \in \mathcal{Z}_{sr}(M)\} \geq 2.$$

Therefore the integrand is strictly larger than one on a set of positive μ_s-measure. Its integral is then strictly larger than $\mu_s(Z)$, which implies

$$\sum_{j \in \mathcal{A}} P_{sj}(M) > 1.$$

□

What is the exact obstruction? The obstruction is overlap. The benchmark definition of $\mathcal{Z}_{sj}(M)$ records all actions that solve the Bellman problem. If the Bellman problem has ties, then the same state appears in more than one solving region. When the transition probabilities are computed by taking the measure of each solving region separately and then adding across columns, the overlapping states are counted more than once. That is why the row-sum-equals-one claim is not automatic.

1.5 Part 4: Supplemental Conditions That Recover a Row-Stochastic Transition Matrix

Why this subsection is explicitly supplemental. The benchmark manuscript wants $\mathbf{P}(M)$ to be a genuine transition matrix. The previous proposition shows that this conclusion needs one more ingredient. We therefore state that ingredient explicitly rather than silently folding it into the benchmark definition.

Assumption 1 (T1: no ties almost everywhere)**.** *For each current type $s \in \{A, B, C\}$,*

$$\sum_{j \in \mathcal{A}} \mathbf{1}\{z \in \mathcal{Z}_{sj}(M)\} = 1 \qquad \textit{for } \mu_s\textit{-almost every } z.$$

Equivalently, the Bellman maximizer is unique for μ_s-almost every state.

Proposition 4 (Row-stochasticity under no ties almost everywhere)**.** *Under Assumption T1, the benchmark transition probabilities satisfy*

$$\sum_{j \in \mathcal{A}} P_{sj}(M) = 1 \qquad \textit{for every current type } s \in \{A, B, C\}.$$

Therefore the benchmark matrix $\mathbf{P}(M)$ is row stochastic.

Proof. Assumption T1 states exactly that

$$\sum_{j \in \mathcal{A}} \mathbf{1}\{z \in \mathcal{Z}_{sj}(M)\} = 1 \qquad \text{for } \mu_s\text{-almost every } z.$$

Integrating both sides with respect to μ_s gives

$$\int_Z \sum_{j \in \mathcal{A}} \mathbf{1}\{z \in \mathcal{Z}_{sj}(M)\}\, \mu_s(dz) = \int_Z 1\, \mu_s(dz) = \mu_s(Z).$$

Using the same indicator-sum identity as in the previous proposition,

$$\sum_{j \in \mathcal{A}} P_{sj}(M) = \frac{1}{\mu_s(Z)} \int_Z \sum_{j \in \mathcal{A}} \mathbf{1}\{z \in \mathcal{Z}_{sj}(M)\}\, \mu_s(dz) = 1.$$

This holds for every current type s, so every row of $\mathbf{P}(M)$ sums to one. □

An explicit selector-based alternative. There is another way to obtain a genuine transition matrix. Instead of assuming ties disappear almost everywhere, one can add an explicit measurable selector

$$g_s(z; M) \in \arg\max_{a \in \mathcal{A}} W_{sa}(z; M).$$

This is exactly the single-valued policy object that later appears in the benchmark stationary-measure law of motion.

Given such a selector, define the selected action regions

$$D_{sj}(M) := \{z \in Z : g_s(z; M) = j\}.$$

By construction, for each current type s the sets $\{D_{sj}(M)\}_{j \in \mathcal{A}}$ are pairwise disjoint and satisfy

$$\bigcup_{j \in \mathcal{A}} D_{sj}(M) = Z.$$

If g_s is measurable, each set $D_{sj}(M)$ is Borel measurable.

Proposition 5 (Selector-based transition matrix)**.** *Suppose there exists a measurable selector g_s for each current type s. Define*

$$P^g_{sj}(M) := \frac{\mu_s\big(D_{sj}(M)\big)}{\mu_s(Z)}.$$

Then for each current type s,

$$\sum_{j\in\mathcal{A}} P^g_{sj}(M) = 1.$$

Proof. Because the selected regions are pairwise disjoint and their union is Z,

$$\sum_{j\in\mathcal{A}} \mu_s\big(D_{sj}(M)\big) = \mu_s\left(\bigcup_{j\in\mathcal{A}} D_{sj}(M)\right) = \mu_s(Z).$$

Dividing by $\mu_s(Z) > 0$ yields

$$\sum_{j\in\mathcal{A}} P^g_{sj}(M) = 1.$$

□

How this maps back to the benchmark. The benchmark object remains the choice-region definition $\mathcal{Z}_{sj}(M)$, because that is how the manuscript first defines the transition statistics. The selector-based construction is an auxiliary implementation device. It becomes relevant only if the later stationary-distribution or simulation block needs a single-valued policy rule that partitions the state space.

1.6 What did we just do, and why does it matter?

We took the value functions and choice-specific values from Phase 3 and converted them into objects that can be compared with transition data. The key step was to define the benchmark choice regions as the states in which each organizational action solves the Bellman problem. Those regions are where dynamic optimization turns into a cross-sectional reorganization pattern.

The measurability proof matters because without measurable choice regions, the expressions $\mu_s(\mathcal{Z}_{sj}(M))$ would not even be legitimate measure-theoretic objects. So measurability is not cosmetic. It is the mathematical condition that makes the transition statistics well defined.

The row-sum issue matters because it clarifies a genuine benchmark subtlety. The choice regions record all maximizing actions. If ties occur, the same state can belong to more than one region. That is harmless when we are only describing the Bellman problem itself, but it becomes consequential once we try to treat the resulting objects as transition probabilities. A probability row must sum to one. Overlapping choice regions can make the raw set-based row sum exceed one.

This is why we separated two statements that are easy to blur together. One statement is benchmark-faithful and unconditional: the choice regions are measurable, and the set-based

ratios are well defined conditional on candidate stationary measures. The second statement requires an extra ingredient: if we want a genuine row-stochastic transition matrix, we need either a no-ties-almost-everywhere condition or an explicit measurable selector. Making that distinction explicit keeps the later stationary-distribution phase logically clean.

1.7 End-of-Section Recap

What has been established mathematically. For fixed (w, p_m, M) and given candidate stationary incumbent measures μ_A, μ_B, μ_C, the benchmark choice regions $\mathcal{Z}_{sj}(M)$ are Borel measurable. Therefore the benchmark ratios $P_{sj}(M) = \mu_s(\mathcal{Z}_{sj}(M))/\mu_s(Z)$ are well defined whenever $\mu_s(Z) > 0$. However, the benchmark set-based formula yields row sum one only if the choice regions partition the state space up to μ_s-null sets. Positive-measure ties make the raw row sum exceed one.

What assumptions were used. We used the fixed-price value-function and choice-specific-value objects established in Phase 3, continuity of the choice-specific values, compactness and measurability of the state space inherited from Phase 1, and finiteness and positivity of the candidate stationary measures. To recover a genuine row-stochastic transition matrix from the benchmark formula, we additionally introduced the supplemental no-ties-almost-everywhere condition T1. As an alternative, we described an explicit measurable selector.

What remains to be derived next. The next phase must define entrant values, entrant masses, and the entrant-composition vector $\mathbf{e}(M)$. After that, the later phases must combine incumbent reorganization, entry, market clearing, and invariant measures into a full stationary equilibrium block.